Block practice · Series and the binomial expansion

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Adding up an arithmetic series between two limits

Writing a sigma sum as a formula in n

Multiplying a binomial expansion by another bracket

Finding r from sums of a geometric series

Finding unknown constants from binomial coefficients

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Practice workbook · Adding up an arithmetic series between two limits

Evaluating a sigma sum that does not start at r = 1 by subtracting two sums from r = 1, or by finding the first term, last term and number of terms.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-05

1. Question 3 · part b · 2 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part b only.

3 (a) Show that $\sum_{r=1}^{n}(5r-3)=\frac{n}{2}(5n-1)$ (3)

(b) Hence, or otherwise, evaluate $\sum_{r=31}^{60}(5r-3)$ (2)

Given that $\sum_{r=1}^{n}(5r-3)=3783$

(c) find the value of n (3)

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1. Question 3 · part b · 2 marks · Working and answer

Working and answer

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2. Question 5 · part b · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part b only.

5 (a) Show that $\sum_{r=1}^{n}(3r+5) = \frac{1}{2}n(3n+13)$ (3)

(b) Hence evaluate $\sum_{r=35}^{50}(3r+5)$ (2)

Given that $\sum_{r=1}^{n}(3r+5) = 385$

(c) find the value of n. (3)

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2. Question 5 · part b · 2 marks · Working and answer

Working and answer

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3. Question 1 · part b · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part b only.

1 (a) Show that $\sum_{r=1}^{n}(3r+2) = \frac{n}{2}(3n+7)$

(3)

(b) Hence, or otherwise, evaluate $\sum_{r=10}^{40}(3r+2)$

(2)

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(Total for Question 1 is 5 marks)

3. Question 1 · part b · 2 marks · Working and answer

Working and answer

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4. Question 9 · part b · 3 marks

2021 NOV · Paper 01

Further practice · Attempt part b only.

9 (a) Show that $\sum_{r=1}^{n}(5r-1)=\frac{n}{2}(3+5n)$ (3)

(b) Hence, or otherwise, evaluate $\sum_{r=10}^{20}(5r-1)$ (3)

The sum of the first n terms of an arithmetic series is S_n where $S_n=\sum_{r=1}^{n}(5r-1)$

The rth term of this series is u_r

Given that $S_n=12u_{n+1}+52$

(c) find the value of n (5)

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4. Question 9 · part b · 3 marks · Working and answer

Working and answer

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5. Question 2 · part b · 4 marks

2024 NOV · Paper 01

Further practice · Attempt part b only.

2 The sum of the fifth, sixth and seventh terms of an arithmetic series A is nine times the sum of the first and second terms.
The third term of A is 12

(a) Find the first term and common difference of A

(5)

The nth term of A is u_n

(b) Find the value of $\sum_{r=15}^{60} u_r$

(4)

The sum to n terms of A is S_n

Given that $2S_n - 5u_n = 10$

(c) find the value of n

(4)

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5. Question 2 · part b · 4 marks · Working and answer

Working and answer

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6. Question 2 · part b · 4 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

2 The nth term of an arithmetic series is u_n

Given that $u_5 = 46$ and that $u_{20} = 181$

(a) find

(i) the common difference of this series,

(ii) the first term of this series.

(4)

(b) Evaluate $\sum_{n=21}^{30} u_n$

(4)

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6. Question 2 · part b · 4 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 2 continued

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(Total for Question 2 is 8 marks)

6. Question 2 · part b · 4 marks · Working and answer

Working and answer

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7. Question 4 · part b.i · 2 marks

2025 NOV · Paper 01

Further practice · Attempt part b.i only.

4 Given that $\sum_{r=1}^{n}(4r+A)=\frac{n}{2}(4n-6)$ where A is a constant

(a) show that $A=-5$

(3)

Hence, or otherwise,

(b) (i) evaluate $\sum_{21}^{60}(4r-5)$

(2)

(ii) find the greatest value of n such that

$$\sum_{r=1}^{n}(4r-5)<3418$$

(3)

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7. Question 4 · part b.i · 2 marks · Working and answer

Working and answer

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Practice workbook · Writing a sigma sum as a formula in n

Recognising that a sum like the sum of (ar + b) from r = 1 to n is arithmetic and writing it as an expression in n.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-04

1. Question 3 · part a · 3 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part a only.

3 (a) Show that $\sum_{r=1}^{n}(5r-3)=\frac{n}{2}(5n-1)$ (3)

(b) Hence, or otherwise, evaluate $\sum_{r=31}^{60}(5r-3)$ (2)

Given that $\sum_{r=1}^{n}(5r-3)=3783$

(c) find the value of n (3)

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1. Question 3 · part a · 3 marks · Working and answer

Working and answer

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2. Question 6 · part a · 3 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

6 (a) Show that $\sum_{r=1}^{n}(4r-3) = n(2n-1)$

(3)

(b) Hence, or otherwise, find the least value of n such that $\sum_{r=1}^{n}(4r-3) > 1000$

(3)

Given that $S_n = n(2n-1)$, $t_n = (4n-3)$ and that $18 + 3t_{n+7} = S_{n+4}$

(c) find the value of n.

(4)

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2. Question 6 · part a · 3 marks · Working and answer

Working and answer

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3. Question 5 · part a · 3 marks

2020 NOV · Paper 02

Further practice · Attempt part a only.

5 (a) Show that $\sum_{r=1}^{n}(3r+5) = \frac{1}{2}n(3n+13)$ (3)

(b) Hence evaluate $\sum_{r=35}^{50}(3r+5)$ (2)

Given that $\sum_{r=1}^{n}(3r+5) = 385$

(c) find the value of n. (3)

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3. Question 5 · part a · 3 marks · Working and answer

Working and answer

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4. Question 1 · part a · 3 marks

2023 JUN · Paper 01

Further practice · Attempt part a only.

1 (a) Show that $\sum_{r=1}^{n}(3r+2) = \frac{n}{2}(3n+7)$

(3)

(b) Hence, or otherwise, evaluate $\sum_{r=10}^{40}(3r+2)$

(2)

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(Total for Question 1 is 5 marks)

4. Question 1 · part a · 3 marks · Working and answer

Working and answer

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5. Question 9 · part a · 3 marks

2021 NOV · Paper 01

Further practice · Attempt part a only.

9 (a) Show that $\sum_{r=1}^{n}(5r-1)=\frac{n}{2}(3+5n)$ (3)

(b) Hence, or otherwise, evaluate $\sum_{r=10}^{20}(5r-1)$ (3)

The sum of the first n terms of an arithmetic series is S_n where $S_n=\sum_{r=1}^{n}(5r-1)$

The rth term of this series is u_r

Given that $S_n=12u_{n+1}+52$

(c) find the value of n (5)

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5. Question 9 · part a · 3 marks · Working and answer

Working and answer

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6. Question 6 · part b · 6 marks

2022 JAN · Paper 02

Further practice · Attempt part b only.

6 The sum of the first n terms of an arithmetic series is S_n where $S_n = n(3 + 2n)$

(a) Find the value of S_{20}

(2)

Given that $S_n = \sum_{r=1}^{n} (Ar + B)$

(b) find the value of A and the value of B

(6)

A different arithmetic series has first term 7 and common difference 4

The sum of the first n terms of this series is T_n

(c) Use algebra to find the value of n for which $T_n = S_n + 252$

(5)

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6. Question 6 · part b · 6 marks · Working and answer

Working and answer

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7. Question 4 · part a · 3 marks

2025 NOV · Paper 01

Further practice · Attempt part a only.

4 Given that $\sum_{r=1}^{n}(4r+A)=\frac{n}{2}(4n-6)$ where A is a constant

(a) show that $A=-5$

(3)

Hence, or otherwise,

(b) (i) evaluate $\sum_{21}^{60}(4r-5)$

(2)

(ii) find the greatest value of n such that

$$\sum_{r=1}^{n}(4r-5)<3418$$

(3)

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7. Question 4 · part a · 3 marks · Working and answer

Working and answer

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Practice workbook · Multiplying a binomial expansion by another bracket

Multiplying an expansion by a linear or polynomial factor and collecting terms up to a given power, sometimes matching to given coefficients.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-21

1. Question 5 · part c · 2 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part c only.

5 Given that $(2 + 3x)^{-1}$ can be expressed in the form $p(1 + qx)^{-1}$ where p and q are constants,

(a) find the value of p and the value of q

(2)

(b) Hence expand $(2 + 3x)^{-1}$ in ascending powers of x up to and including the term in x^3, expressing each coefficient as an exact fraction in its lowest terms.

(3)

$$f(x) = \frac{1 + x}{2 + 3x}$$

(c) Obtain a series expansion for $f(x)$, in ascending powers of x up to and including the term in x^3, expressing each coefficient as an exact fraction in its lowest terms.

(2)

(d) Hence use algebraic integration to obtain an estimate, to 4 decimal places, of

$$\int_0^{0.5} f(x)\,dx$$

(4)

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1. Question 5 · part c · 2 marks · Working and answer

Working and answer

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2. Question 9 · part c · 3 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

9 (a) Expand $(1+2x)^{-\frac{1}{3}}$ in ascending powers of x up to and including the term in x^3 expressing each coefficient as a fraction in its lowest terms.

(3)

(b) Find the range of values of x for which your expansion is valid.

(1)

$$f(x) = \frac{2+kx^2}{(1+2x)^{\frac{1}{3}}}$$

(c) Obtain a series expansion of f (x) in ascending powers of x up to and including the term in x^3
Give your coefficients in terms of k where appropriate.

(3)

The coefficient of x^3 in the series expansion of f (x) is $-\frac{8}{3}$

(d) Find the exact value of k

(2)

(e) Hence, using algebraic integration, estimate the value of

$$\int_{0.1}^{0.2} f(x)\,dx$$

Give your answer to 4 decimal places.

(5)

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2. Question 9 · part c · 3 marks · Working and answer

Working and answer

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3. Question 7 · part b · 2 marks

2024 JUN · Paper 02

Further practice · Attempt part b only.

7 (a) Expand $\left(1+2x^2\right)^{-\frac{3}{4}}$ in ascending powers of x up to and including the term in x^6

Express each coefficient as an exact fraction in its lowest terms.

(3)

$$f(x) = \frac{(2+kx)}{\left(1+2x^2\right)^{\frac{3}{4}}} \quad \text{where } k \neq 0$$

(b) Obtain a series expansion for $f(x)$ in ascending powers of x up to and including the term in x^5

Give each coefficient in terms of k where appropriate.

(2)

The coefficient of the term in x^5 is fourteen times the coefficient of the term in x^2

(c) Find the value of k

(2)

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3. Question 7 · part b · 2 marks · Working and answer

Working and answer

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4. Question 10 · part c · 4 marks

2021 NOV · Paper 02

Further practice · Attempt part c only.

10 (a) Show that $\frac{3}{\sqrt{9-3x}} = \left(1-\frac{x}{3}\right)^{-\frac{1}{2}}$

(2)

(b) Hence expand $\frac{3}{\sqrt{9-3x}}$ in ascending powers of x up to and including the term in x^3 expressing each coefficient as an exact fraction in its lowest terms.

(3)

$$f(x) = \frac{1+2x}{\sqrt{9-3x}}$$

(c) Find the expansion of $3f(x)$ in ascending powers of x up to and including the term in x^3 expressing each coefficient as an exact fraction in its lowest terms.

(4)

(d) Hence, using algebraic integration, obtain an approximation to 6 significant figures for

$$\int_{0.1}^{0.2} \frac{1+2x}{\sqrt{9-3x}}\,dx$$

(4)

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4. Question 10 · part c · 4 marks · Working and answer

Working and answer

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5. Question 7 · part d · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part d only.

7 (a) Expand $\left(1+\frac{x}{3}\right)^{-3}$ in ascending powers of x up to and including the term in x^3

Where appropriate express each coefficient as an exact fraction in its lowest terms.

(3)

(b) Write down the range of values of x for which your expression is valid.

(1)

(c) Express $(3+x)^{-3}$ in the form $P(1+Qx)^{-3}$ where P and Q are rational numbers whose values should be stated.

(2)

$$f(x)=\frac{(1+4x)}{(3+x)^3}$$

(d) Obtain a series expansion for $f(x)$ in ascending powers of x up to and including the term in x^2

(2)

(e) Hence, using algebraic integration, obtain an estimate of $\int_0^{0.2} f(x)\,dx$

Give your answer to 5 significant figures.

(3)

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5. Question 7 · part d · 2 marks · Working and answer

Working and answer

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6. Question 9 · part c · 3 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

9 Given that $\frac{1}{(2-x)^3}$ can be written as $p(1-qx)^{-3}$

(a) find the value of p and the value of q.

(2)

(b) Expand $\frac{1}{(2-x)^3}$ in ascending powers of x up to and including the term in x^3 and express each coefficient as an exact fraction in its lowest terms.

(3)

$$f(x) = \frac{a+bx}{(2-x)^3} \text{ where } a \text{ and } b \text{ are integers}$$

The first three terms of the expansion of $f(x)$ are $\frac{3}{8} - \frac{43}{16}x + cx^2$

(c) Find the value of a and the value of b.

(3)

(d) Find the exact value of c.

(2)

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6. Question 9 · part c · 3 marks · Working and answer

Working and answer

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7. Question 9 · part d · 2 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

9 Given that $\frac{1}{(2-x)^3}$ can be written as $p(1-qx)^{-3}$

(a) find the value of p and the value of q.

(2)

(b) Expand $\frac{1}{(2-x)^3}$ in ascending powers of x up to and including the term in x^3 and express each coefficient as an exact fraction in its lowest terms.

(3)

$$f(x) = \frac{a+bx}{(2-x)^3} \text{ where } a \text{ and } b \text{ are integers}$$

The first three terms of the expansion of $f(x)$ are $\frac{3}{8} - \frac{43}{16}x + cx^2$

(c) Find the value of a and the value of b.

(3)

(d) Find the exact value of c.

(2)

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7. Question 9 · part d · 2 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 9 continued

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7. Question 9 · part d · 2 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 9 continued

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7. Question 9 · part d · 2 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.



Question 9 continued

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(Total for Question 9 is 10 marks)

7. Question 9 · part d · 2 marks · Working and answer

Working and answer

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8. Question 9 · part b · 4 marks

2023 NOV · Paper 01

Further practice · Attempt part b only.

9 (a) Expand $\left(1-8x^2\right)^{\frac{1}{2}}$ in ascending powers of x, up to and including the term in x^6 giving each coefficient as an integer.

(3)

$$g(x)=\frac{a+bx}{\sqrt{1-8x^2}} \quad \text{where } a \text{ and } b \text{ are prime numbers}$$

Given that the fourth and fifth terms, in ascending powers of x, in the series expansion of $g(x)$ are $20x^3$ and $48x^4$ respectively,

(b) find the value of a and the value of b

(4)

Using the first five terms, in ascending powers of x, in the series expansion of $g(x)$

(c) obtain an estimate, to 4 significant figures, of $\int_0^{0.2} g(x)\,dx$

(4)

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8. Question 9 · part b · 4 marks · Working and answer

Working and answer

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Practice workbook · Finding r from sums of a geometric series

Writing given sums (and terms) of a geometric series in a and r, eliminating a, and solving the resulting quadratic or cubic in r.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-10

1. Question 8 · part a · 4 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part a only.

8 The sum of the first and second terms of a geometric series G is 400

The sum of the second and third terms of G is 100

(a) Show that the common ratio of G is $\frac{1}{4}$

(4)

(b) Show that the first term of G is 320

(2)

(c) Find the sum to infinity of G

(2)

The sum to n terms of G is S_n

(d) Find, using logarithms, the least value of n such that

$$S_n > 426.6$$

(4)

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1. Question 8 · part a · 4 marks · Working and answer

Working and answer

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2. Question 5 · part a · 5 marks

2021 JUN · Paper 02

Further practice · Attempt part a only.

5 The nth term of a geometric series with common ratio r is u_n

Given that $u_2 + u_4 = 212.5$ and that $u_3 + u_4 = 62.5$

(a) find the two possible values of r.

(5)

Given that the series is convergent with sum to infinity S,

(b) find the exact value of S.

(2)

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2. Question 5 · part a · 5 marks · Working and answer

Working and answer

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3. Question 7 · part b · 5 marks

2020 JAN · Paper 02

Further practice · Attempt part b only.

7 The 7th term of a geometric series is 192 and the 8th term of this geometric series is 1152

(a) Find, as a fraction in its simplest form, the 4th term of this geometric series.

(3)

A different geometric series G has a common ratio r and nth term t_n

Given that $t_3 = 24$ and $t_2 + t_3 + t_4 = -36$

(b) show that r satisfies the equation

$$2r^2 + 5r + 2 = 0$$

(5)

Given further that G is convergent with sum to infinity S,

(c) find the value of S.

(4)

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3. Question 7 · part b · 5 marks · Working and answer

Working and answer

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4. Question 8 · part b · 2 marks

2024 JUN · Paper 01

Further practice · Attempt part b only.

8 The sum of the first and second terms of a geometric series G is 400

The sum of the second and third terms of G is 100

(a) Show that the common ratio of G is $\frac{1}{4}$

(4)

(b) Show that the first term of G is 320

(2)

(c) Find the sum to infinity of G

(2)

The sum to n terms of G is S_n

(d) Find, using logarithms, the least value of n such that

$$S_n > 426.6$$

(4)

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4. Question 8 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 8 · part a · 7 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

8 The sum of the first 2 terms of a geometric series G is 360
The sum of the 2nd and 3rd terms of G is 288

The nth term of G is U_n

(a) Show that $U_n = A\left(\frac{4}{5}\right)^{n-1}$ where A is an integer to be found.

(7)

(b) Explain why G is convergent.

(1)

(c) Hence find the sum to infinity of G

(2)

(d) Find the least number of terms for which the sum is greater than 978

(4)

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5. Question 8 · part a · 7 marks · Working and answer

Working and answer

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6. Question 9 · part a · 8 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

9 A geometric series G has common ratio r where $r > 0$

The third term of G is $\frac{27}{2}$ and the sum of the first three terms of G is $\frac{57}{2}$

Given that the sum to n terms of G is S_n

(a) show that $S_n = \sum_{j=1}^{n} 4\left(\frac{3}{2}\right)^j$

(8)

Given that $S_k > 50\,000$

(b) show that the least value of k is given by

$$k > \frac{\lg\left(\frac{12\,503}{3}\right)}{\lg\left(\frac{3}{2}\right)}$$

(3)

(c) Hence find the least value of k

(1)

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6. Question 9 · part a · 8 marks · Working and answer

Working and answer

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7. Question 7 · part a.i · 0 marks

2021 NOV · Paper 02

Further practice · Attempt part a.i only.

7 A geometric series G has first term a and common ratio r

The sum of the first three terms of G is $\frac{61}{6}$

The sum to infinity of G is $\frac{125}{6}$

(a) (i) Show that $r = \frac{4}{5}$

(ii) Find the value of a

(6)

The sum of the first n terms of G is S_n

Given that $S_n > 19.8$

(b) show that $n \lg\left(\frac{4}{5}\right) < \lg\left(\frac{31}{625}\right)$

(2)

(c) Hence find the least value of n

(2)

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7. Question 7 · part a.i · 0 marks · Working and answer

Working and answer

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8. Question 7 · part a.ii · 6 marks

2021 NOV · Paper 02

Further practice · Attempt part a.ii only.

7 A geometric series G has first term a and common ratio r

The sum of the first three terms of G is $\frac{61}{6}$

The sum to infinity of G is $\frac{125}{6}$

(a) (i) Show that $r = \frac{4}{5}$

(ii) Find the value of a

(6)

The sum of the first n terms of G is S_n

Given that $S_n > 19.8$

(b) show that $n \lg\left(\frac{4}{5}\right) < \lg\left(\frac{31}{625}\right)$

(2)

(c) Hence find the least value of n

(2)

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8. Question 7 · part a.ii · 6 marks · Working and answer

Working and answer

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9. Question 7 · part c · 3 marks

2024 NOV · Paper 02

Further practice · Attempt part c only.

7 (a) Use the factor theorem to show that $(4x-1)$ is a factor of

$$f(x) = 64x^3 - 64x^2 + 3$$

(2)

(b) Hence, or otherwise, find the exact roots of the equation

$$f(x) = 0$$

(4)

A geometric series G has first term a and common ratio r
The third term of G is 9 and the sum to infinity of G is 192

(c) Show that $64r^3 - 64r^2 + 3 = 0$

(3)

Given that r is a rational number

(d) write down the value of r

(1)

(e) show that $a = 144$

(2)

The sum to n terms of G is S_n

(f) Using logarithms, find the least value of n such that $S_n > 191.9$

(4)

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9. Question 7 · part c · 3 marks · Working and answer

Working and answer

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10. Question 7 · part d · 1 mark

2024 NOV · Paper 02

Further practice · Attempt part d only.

7 (a) Use the factor theorem to show that $(4x-1)$ is a factor of

$$f(x) = 64x^3 - 64x^2 + 3$$

(2)

(b) Hence, or otherwise, find the exact roots of the equation

$$f(x) = 0$$

(4)

A geometric series G has first term a and common ratio r
The third term of G is 9 and the sum to infinity of G is 192

(c) Show that $64r^3 - 64r^2 + 3 = 0$

(3)

Given that r is a rational number

(d) write down the value of r

(1)

(e) show that $a = 144$

(2)

The sum to n terms of G is S_n

(f) Using logarithms, find the least value of n such that $S_n > 191.9$

(4)

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10. Question 7 · part d · 1 mark · Working and answer

Working and answer

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11. Question 7 · part e · 2 marks

2024 NOV · Paper 02

Further practice · Attempt part e only.

7 (a) Use the factor theorem to show that $(4x-1)$ is a factor of

$$f(x) = 64x^3 - 64x^2 + 3$$

(2)

(b) Hence, or otherwise, find the exact roots of the equation

$$f(x) = 0$$

(4)

A geometric series G has first term a and common ratio r
The third term of G is 9 and the sum to infinity of G is 192

(c) Show that $64r^3 - 64r^2 + 3 = 0$

(3)

Given that r is a rational number

(d) write down the value of r

(1)

(e) show that $a = 144$

(2)

The sum to n terms of G is S_n

(f) Using logarithms, find the least value of n such that $S_n > 191.9$

(4)

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11. Question 7 · part e · 2 marks · Working and answer

Working and answer

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12. Question 4 · part whole · 7 marks

2022 JUN · Paper 01

Further practice · Attempt part whole only.

4 The common ratio of a geometric series G is positive.

The sum of the first 4 terms of G is 80
The sum to infinity of G is 81

Show that the sum of the first 7 terms of G differs from the sum to infinity of G by $\frac{1}{27}$

(7)

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12. Question 4 · part whole · 7 marks

2022 JUN · Paper 01

Further practice · Attempt part whole only.

Question 4 continued

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(Total for Question 4 is 7 marks)

12. Question 4 · part whole · 7 marks · Working and answer

Working and answer

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13. Question 7 · part a · 2 marks

2023 NOV · Paper 01

Further practice · Attempt part a only.

7 A geometric series G with common ratio r, has first term 16 and third term $\frac{2704}{625}$

(a) Find the two possible values of r

(2)

Given that $r > 0$

(b) find the sum to infinity of G

(2)

The sum to n terms of G is greater than 33

(c) Find, using logarithms, the least possible value of n
Show your working clearly.

(5)

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13. Question 7 · part a · 2 marks · Working and answer

Working and answer

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Practice workbook · Finding unknown constants from binomial coefficients

Writing coefficients of a binomial expansion in terms of unknown constants (such as n and a) and using given information to solve for them.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-22

1. Question 5 · part b · 6 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part b only.

5 (a) Expand $(1 + ax)^n$ in ascending powers of x up to and including the term in x^3

Express each coefficient of x in terms of a and n where a and n are constants and $n > 2$

(2)

The coefficient of x is 15 and the coefficient of x^2 is equal to the coefficient of x^3

(b) Find the value of a and the value of n.

(6)

(c) Find the coefficient of x^3

(2)

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1. Question 5 · part b · 6 marks · Working and answer

Working and answer

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2. Question 2 · part a · 6 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

2 Given that

$$1 - \frac{1}{3}x + \frac{5}{36}x^2 + \ldots$$

is the binomial expansion, in ascending powers of x, of $(1 + Ax)^n$

where A and n are rational numbers,

(a) find the value of A and the value of n

(6)

(b) Hence find the value of the coefficient of x^3

Give your answer in the form $-\frac{p}{q}$ where p is a prime number and q is an integer.

(2)

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2. Question 2 · part a · 6 marks · Working and answer

Working and answer

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3. Question 2 · part b · 2 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

2 Given that

$$1 - \frac{1}{3}x + \frac{5}{36}x^2 + \ldots$$

is the binomial expansion, in ascending powers of x, of $(1 + Ax)^n$

where A and n are rational numbers,

(a) find the value of A and the value of n

(6)

(b) Hence find the value of the coefficient of x^3

Give your answer in the form $-\frac{p}{q}$ where p is a prime number and q is an integer.

(2)

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3. Question 2 · part b · 2 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 2 continued

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(Total for Question 2 is 8 marks)

3. Question 2 · part b · 2 marks · Working and answer

Working and answer

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4. Question 5 · part a · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part a only.

5 (a) Expand $(1 + ax)^n$ in ascending powers of x up to and including the term in x^3

Express each coefficient of x in terms of a and n where a and n are constants and $n > 2$

(2)

The coefficient of x is 15 and the coefficient of x^2 is equal to the coefficient of x^3

(b) Find the value of a and the value of n.

(6)

(c) Find the coefficient of x^3

(2)

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4. Question 5 · part a · 2 marks · Working and answer

Working and answer

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5. Question 5 · part c · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

5 (a) Expand $(1 + ax)^n$ in ascending powers of x up to and including the term in x^3

Express each coefficient of x in terms of a and n where a and n are constants and $n > 2$

(2)

The coefficient of x is 15 and the coefficient of x^2 is equal to the coefficient of x^3

(b) Find the value of a and the value of n.

(6)

(c) Find the coefficient of x^3

(2)

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5. Question 5 · part c · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

Question 5 continued

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(Total for Question 5 is 10 marks)

5. Question 5 · part c · 2 marks · Working and answer

Working and answer

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6. Question 5 · part a · 2 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

5

$$f(x) = (1 + ax)^n \text{ where } n > 2$$

(a) Write down the expansion of $f(x)$ in ascending powers of x up to and including the term in x^3
Give each coefficient in terms of n and a

(2)

In the expansion of $f(x)$, the coefficient of x is 8 and the coefficient of x^2 is 30

(b) Find the value of n and the value of a

(4)

(c) Calculate the coefficient of x^3 in $f(x)$

(2)

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6. Question 5 · part a · 2 marks · Working and answer

Working and answer

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7. Question 5 · part b · 4 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

5

$$f(x) = (1 + ax)^n \text{ where } n > 2$$

(a) Write down the expansion of $f(x)$ in ascending powers of x up to and including the term in x^3
Give each coefficient in terms of n and a

(2)

In the expansion of $f(x)$, the coefficient of x is 8 and the coefficient of x^2 is 30

(b) Find the value of n and the value of a

(4)

(c) Calculate the coefficient of x^3 in $f(x)$

(2)

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7. Question 5 · part b · 4 marks · Working and answer

Working and answer

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8. Question 5 · part c · 2 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

5

$$f(x) = (1 + ax)^n \text{ where } n > 2$$

(a) Write down the expansion of $f(x)$ in ascending powers of x up to and including the term in x^3
Give each coefficient in terms of n and a

(2)

In the expansion of $f(x)$, the coefficient of x is 8 and the coefficient of x^2 is 30

(b) Find the value of n and the value of a

(4)

(c) Calculate the coefficient of x^3 in $f(x)$

(2)

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8. Question 5 · part c · 2 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 5 continued

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(Total for Question 5 is 8 marks)

8. Question 5 · part c · 2 marks · Working and answer

Working and answer

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9. Question 6 · part b · 4 marks

2024 NOV · Paper 01

Further practice · Attempt part b only.

6 (a) Show that

$$\frac{a}{\sqrt{4+bx}} = \frac{a}{2}\left(1+\frac{bx}{4}\right)^{-\frac{1}{2}}$$ where a and b are positive integers.

(2)

The expansion of $\frac{a}{\sqrt{4+bx}}$ in ascending powers of x can be written as

$$P+Qx+Rx^2+Sx^3$$

where P, Q, R and S are rational numbers.

(b) Show that $Q=-\frac{ab}{16}$ and $S=-\frac{5ab^3}{2048}$

and find P and R in terms of a and b, as fractions in their lowest terms.

(4)

Given that $Q=\frac{128}{5}S$ and $R=\frac{9}{256}$

(c) show that $a=3$ and $b=1$

(3)

(d) Hence, using an appropriate value of x, find, to 3 decimal places, an approximate value for $\frac{\sqrt{6}}{2}$

(3)

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9. Question 6 · part b · 4 marks · Working and answer

Working and answer

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10. Question 6 · part c · 3 marks

2024 NOV · Paper 01

Further practice · Attempt part c only.

6 (a) Show that

$$\frac{a}{\sqrt{4+bx}} = \frac{a}{2}\left(1+\frac{bx}{4}\right)^{-\frac{1}{2}}$$ where a and b are positive integers.

(2)

The expansion of $\frac{a}{\sqrt{4+bx}}$ in ascending powers of x can be written as

$$P+Qx+Rx^2+Sx^3$$

where P, Q, R and S are rational numbers.

(b) Show that $Q=-\frac{ab}{16}$ and $S=-\frac{5ab^3}{2048}$

and find P and R in terms of a and b, as fractions in their lowest terms.

(4)

Given that $Q=\frac{128}{5}S$ and $R=\frac{9}{256}$

(c) show that $a=3$ and $b=1$

(3)

(d) Hence, using an appropriate value of x, find, to 3 decimal places, an approximate value for $\frac{\sqrt{6}}{2}$

(3)

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10. Question 6 · part c · 3 marks · Working and answer

Working and answer

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11. Question 3 · part b · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part b only.

3 (a) Expand

$$(1 + px)^{-5} \quad p \neq 0$$

in ascending powers of x, up to and including the term in x^4
Give each term in its simplest form.

(3)

The coefficient of x^r in the expansion is c_r

Given that $c_4 = 2c_3$

(b) find the value of p.

(2)

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11. Question 3 · part b · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part b only.

Question 3 continued

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(Total for Question 3 is 5 marks)

11. Question 3 · part b · 2 marks · Working and answer

Working and answer

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12. Question 9 · part d · 2 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

9 (a) Expand $(1 + 2x)^{-\frac{1}{3}}$ in ascending powers of x up to and including the term in x^3 expressing each coefficient as a fraction in its lowest terms.

(3)

(b) Find the range of values of x for which your expansion is valid.

(1)

$$f(x) = \frac{2 + kx^2}{(1 + 2x)^{\frac{1}{3}}}$$

(c) Obtain a series expansion of f (x) in ascending powers of x up to and including the term in x^3
Give your coefficients in terms of k where appropriate.

(3)

The coefficient of x^3 in the series expansion of f (x) is $-\frac{8}{3}$

(d) Find the exact value of k

(2)

(e) Hence, using algebraic integration, estimate the value of

$$\int_{0.1}^{0.2} f(x)\,dx$$

Give your answer to 4 decimal places.

(5)

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12. Question 9 · part d · 2 marks · Working and answer

Working and answer

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13. Question 7 · part c · 2 marks

2024 JUN · Paper 02

Further practice · Attempt part c only.

7 (a) Expand $\left(1+2x^2\right)^{-\frac{3}{4}}$ in ascending powers of x up to and including the term in x^6

Express each coefficient as an exact fraction in its lowest terms.

(3)

$$f(x) = \frac{(2+kx)}{\left(1+2x^2\right)^{\frac{3}{4}}} \quad \text{where } k \neq 0$$

(b) Obtain a series expansion for $f(x)$ in ascending powers of x up to and including the term in x^5

Give each coefficient in terms of k where appropriate.

(2)

The coefficient of the term in x^5 is fourteen times the coefficient of the term in x^2

(c) Find the value of k

(2)

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13. Question 7 · part c · 2 marks

2024 JUN · Paper 02

Further practice · Attempt part c only.

Question 7 continued

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13. Question 7 · part c · 2 marks

2024 JUN · Paper 02

Further practice · Attempt part c only.

Question 7 continued

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13. Question 7 · part c · 2 marks

2024 JUN · Paper 02

Further practice · Attempt part c only.

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Question 7 continued

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(Total for Question 7 is 7 marks)

13. Question 7 · part c · 2 marks · Working and answer

Working and answer

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